Zatomic

ZECATHON · CROSS-CHAIN · P2P OTC

Swap shielded ZEC for ETH.

No bridge. No wrapped ZEC.

Both exits are prepared before funding. If the trade fails, Alice recovers her ZEC through **a pre-signed, Zcash-enforced refund**, without further help from Bob.

THE PROBLEM

Why a shielded atomic swap is hard

01

Neither side wants to go first

Whoever pays first can be cheated, so both payments must happen, or neither.

02

Shielded notes have no scripts

No hash locks or time locks: a shielded note moves only with a valid signature from its key.

03

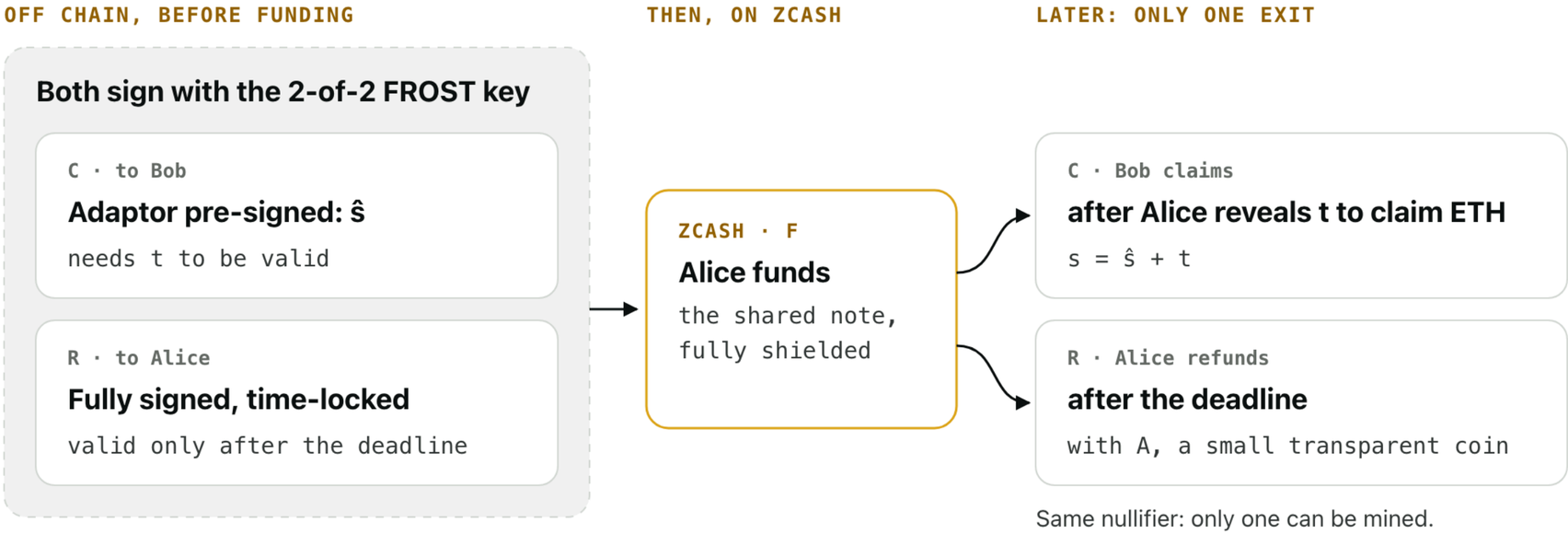
Other routes make different trade-offs

Some designs require cross-chain verification; others introduce intermediaries or privacy trade-offs.

> Zatomic keeps the ZEC shielded and asks Ethereum to verify **no Zcash history**.

WHAT IS NEW · PREPARED BEFORE FUNDING

One note, two exits, both signed first



Signed before the note exists: in Zcash v6, signatures do not cover the anchor or the proof. **Alice's refund needs nothing from Bob and no proof from Ethereum**: Zcash enforces its lock time.

A settled swap stays fully shielded. Only a refund uses one small transparent coin from Alice's own change.

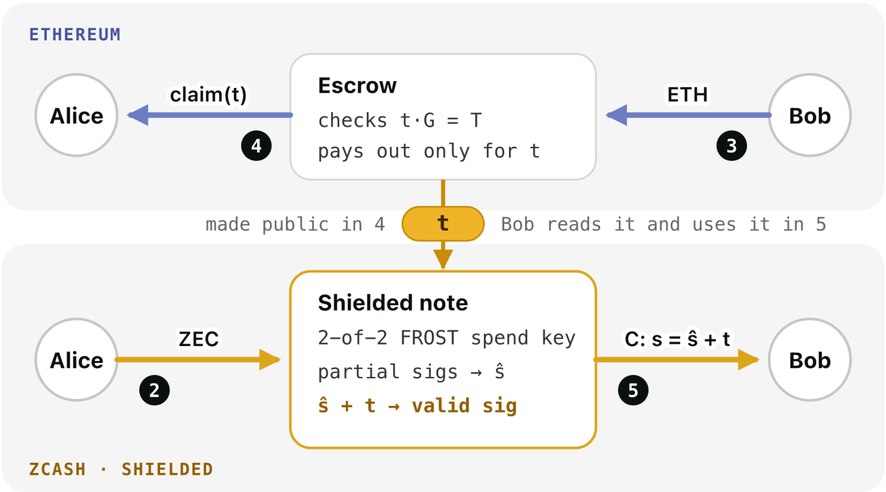
HOW IT WORKS

Two payments, one secret

1 **Off chain, before funds move:**
both sign the two exits in advance with the 2-of-2 FROST key

Exit 1: C pays the ZEC to Bob. Their joint signature ŝ needs t to be valid.

Exit 2: R returns it to Alice. Fully signed, but not valid until a deadline.



IF NOBODY REVEALS t

After the deadlines: Bob takes his ETH back, Alice's refund returns her ZEC.

1 Off chain, before any funds move: both exits are signed.

2 Alice locks ZEC in a note whose spend key is a 2-of-2 FROST key.

3 Bob locks ETH in the escrow, bound to T = t·G.

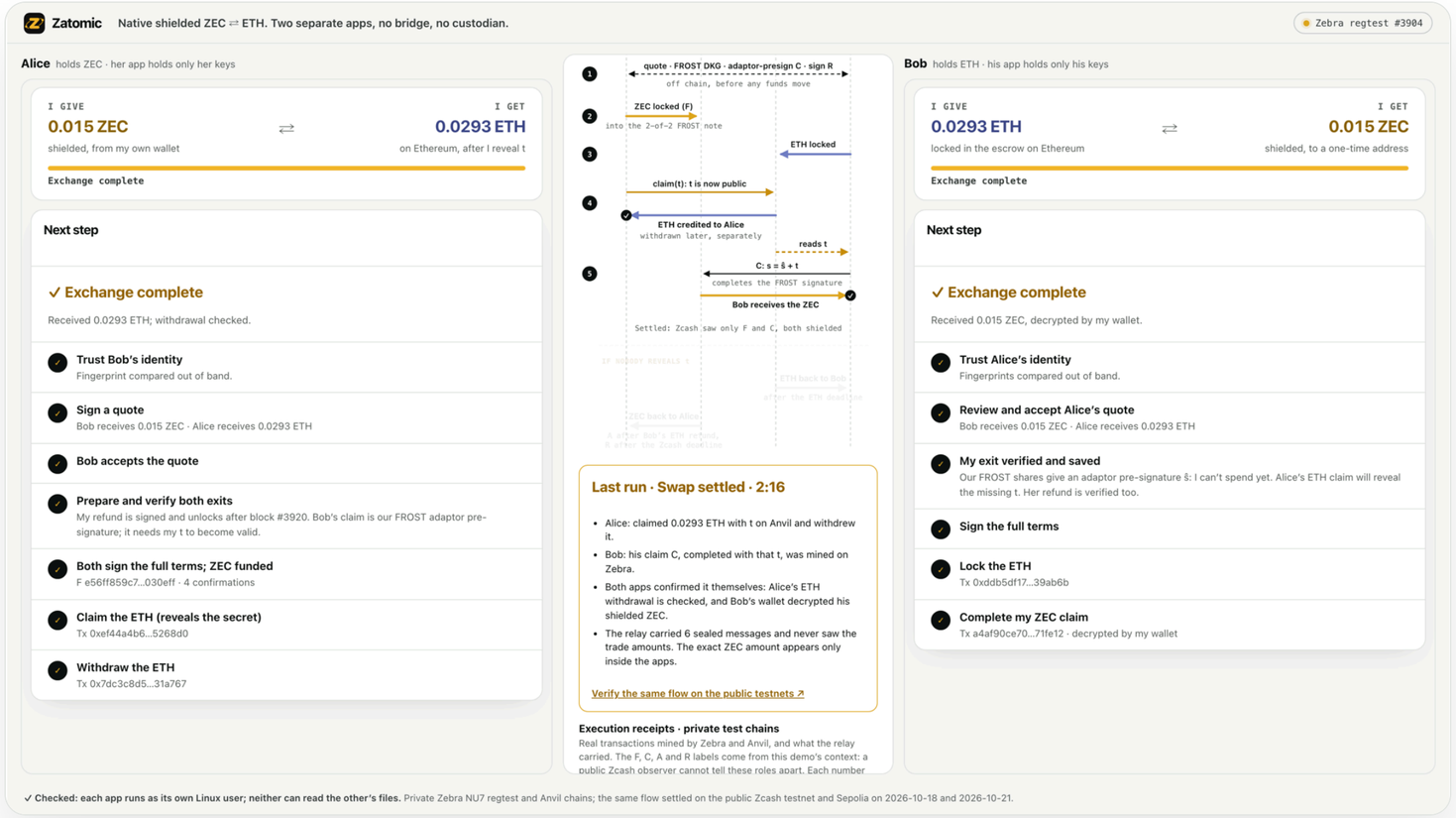
4 Alice takes the ETH by revealing t; the escrow checks t·G = T on Pallas.

> **5** The same t completes their FROST adaptor pre-signature: **s = ŝ + t**.

LIVE DEMO

Two independent owner apps. A relay that cannot read their trade.

Each app holds only its owner’s keys. Real Zebra NU7 and Anvil transactions on private test chains, each with its block and ID.



PUBLIC TESTNETS · ZCASH TESTNET + SEPOLIA · 2026-10-21

Both outcomes, on real networks

SUCCESS PATH

Alice receives the ETH

Bob receives shielded ZEC

`Zcash: F + C, fully shielded`

NO-REVEAL PATH

Bob recovers his ETH

Alice recovers shielded ZEC

`Zcash: F (shielded) → A → time-locked R`

Run on 2026-10-21 with the shipped V2 escrow (295,236 gas per settled swap, −79% vs V1). Every transaction is linked in the records.

`Live demo · 162-55-186-88.sslip.io` `Code · github.com/yugocabrio/zatomic`